EXCELLENCE

CLASS 11 - 2026-27

Physics | Paper Discussion

Basic Maths and Calculus

Weekly Test 02 · Lectures 01–10

By - Excellence Programme

20 September 2026

How we will discuss the paper

01 30 MCQs from Weekly Test 02 (Lectures 01–10)

02 Question on one slide → solution on the next

03 Same methods we used in class (ASTC, $\sqrt{(a^2+b^2)}$, $Tn = Sn - S_{n-1}$, ...)

04 Keep steps short and easy — board style

05 Rough-work slides appear every few questions

Lock lines before we start

ASTC: sign first · $a\sin\theta + b\cos\theta \rightarrow \sqrt{(a^2+b^2)}$ · arc = $r\theta$ (radian)

Triangle ratios from the right triangle. Identities: $(\sec - \tan)(\sec + \tan) = 1$.

AP: $T_n = S_n - S_{n-1}$. GP rebound: first fall + 2 × infinite series of ups.

Circle: centre $(-g/2, -f/2)$. Ellipse foci: $2ae$. Diff: product / quotient / chain.

Do not invent a new method on the paper. Use the same board method from class.

Q1. L01 ratios

In △ABC, ∠B = 90°. If tan A = 5/12, then sin A + cos A equals

2 marks

A 17/13

B 17/12

C 7/13

D 12/5

Solution: Q1. L01 ratios

Step 1: tan A = opp/adj = 5/12 → draw the right triangle

Step 2: hyp = √(5² + 12²) = √(25 + 144) = √169 = 13

Step 3: sin A = 5/13, cos A = 12/13

Step 4: sin A + cos A = 5/13 + 12/13 = 17/13

Answer: A 17/13

Do not forget to find the hypotenuse with Pythagoras first.

Rough work / extra work

Q2. L01 identities

If $\sec\theta - \tan\theta = 1/2$, then $\sec\theta + \tan\theta$ equals

2 marks

A 1/2

B 1

C 2

D 4

Solution: Q2. L01 identities

$$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$$

Step 1: Use the identity $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = \sec^2\theta - \tan^2\theta = 1$

Step 2: So $(\sec\theta + \tan\theta) = 1 / (\sec\theta - \tan\theta)$

Step 3: $\sec\theta + \tan\theta = 1 / (1/2) = 2$

Answer: C 2

Q3. L01 identities

If $\sin\theta + \cos\theta = \sqrt{2}$, then $\tan\theta + \cot\theta$ equals

2 marks

A 1

B 2

C $\sqrt{2}$

D 0

Solution: Q3. L01 identities

Step 1: Square both sides: $(\sin\theta + \cos\theta)^2 = 2$

Step 2: $1 + 2\sin\theta\cos\theta = 2 \rightarrow \sin 2\theta = 1 \rightarrow 2\theta = 90°$ (acute) $\rightarrow \theta = 45° = \pi/4$

Step 3: $\tan(\pi/4) = 1$, $\cot(\pi/4) = 1$

Step 4: $\tan\theta + \cot\theta = 1 + 1 = 2$

Answer: B 2

After squaring, take the acute angle that fits $0 < \theta < \pi/2$.

Q4. L02 ASTC

The value of $\sin 510° \cdot \cos 330° - \cos 390° \cdot \sin 150°$ is

2 marks

A 0

B $1/2$

C $\sqrt{3}/2$

D $-1/2$

Solution: Q4. L02 ASTC

Step 1: Reduce each angle: $\sin 510° = \sin(510 - 360) = \sin 150° = 1/2$

Step 2: $\cos 330° = +\cos 30° = \sqrt{3}/2$ (ASTC: 4th quadrant, cos +)

Step 3: $\cos 390° = \cos 30° = \sqrt{3}/2$

Step 4: Value = $(1/2)(\sqrt{3}/2) - (\sqrt{3}/2)(1/2) = 0$

Answer: A 0

Sign from ASTC first, then the acute brother angle.

Rough work / extra work

Q5. L02 ASTC

The value of $\cos(-1410°)$ is

2 marks

A $1/2$

B $-1/2$

C $\sqrt{3}/2$

D $-\sqrt{3}/2$

Solution: Q5. L02 ASTC

$$\cos(-\theta) = \cos\theta$$

Step 1: cos is even: $\cos(-1410°) = \cos 1410°$

Step 2: $1410° = 3 \times 360° + 330° \rightarrow \cos 1410° = \cos 330°$

Step 3: $\cos 330° = +\cos 30° = \sqrt{3}/2$ (ASTC 4th quadrant)

Answer: C $\sqrt{3}/2$

Q6. L02 radian

An arc of a circle of radius 21 cm subtends 60° at the centre. The length of the arc is

2 marks

A 7π cm

B 11π cm

C 14π cm

D 22π cm

Solution: Q6. L02 radian

$\ell = r\theta$ (θ in radian)

Step 1: Convert to radian: $60° = 60 \times \pi/180 = \pi/3$

Step 2: Arc length $\ell = r\theta = 21 \times (\pi/3) = 7\pi$ cm

Answer: A 7π cm

Never put degrees straight into $\ell = r\theta$.

Rough work / extra work

Q7. L02 height

The angle of elevation of a tower from a point A is 30°. From a point B, 20 m closer to the tower on the same horizontal line, the elevation is 60°. The height of the tower is

2 marks

A $10\sqrt{3}$ m

B $20\sqrt{3}$ m

C 10 m

D 20 m

Solution: Q7. L02 height

Step 1: Let height = h, and distance from A to base = x

Step 2: From A: $\tan 30° = h/x \rightarrow 1/\sqrt{3} = h/x \rightarrow x = h\sqrt{3}$

Step 3: From B $(x - 20)$: $\tan 60° = h/(x - 20) \rightarrow \sqrt{3} = h/(x - 20)$

Step 4: $h\sqrt{3} - 20 = h/\sqrt{3}$ → multiply by $\sqrt{3}$: $3h - 20\sqrt{3} = h \rightarrow 2h = 20\sqrt{3} \rightarrow h = 10\sqrt{3}$

Answer: A $10\sqrt{3}$ m

Q8. L03 compound

If $\sin(A + B) = 1$ and $\cos(A - B) = \sqrt{3}/2$ with $0 < A, B < \pi/2$, then A equals

2 marks

A $\pi/3$

B $\pi/4$

C $\pi/6$

D $\pi/2$

Solution: Q8. L03 compound

Step 1: $\sin(A + B) = 1 \rightarrow A + B = \pi/2$

Step 2: $\cos(A - B) = \sqrt{3}/2 \rightarrow A - B = \pi/6$ (acute)

Step 3: Add the two equations: $2A = \pi/2 + \pi/6 = 2\pi/3$

Step 4: $A = \pi/3$

Answer: A $\pi/3$

Q9. L03 double

If $\tan\theta = 1/2$, then $(2\tan\theta)/(1 - \tan^2\theta)$ equals

2 marks

A 4/3

B 3/4

C 1

D 4

Solution: Q9. L03 double

$$\tan 2\theta = (2 \tan \theta)/(1 - \tan^2 \theta)$$

Step 1: Recognise the double-angle formula: $\tan 2\theta = (2 \tan \theta)/(1 - \tan^2 \theta)$

Step 2: Put $\tan \theta = 1/2$: numerator $= 2 \times (1/2) = 1$

Step 3: denominator $= 1 - (1/2)^2 = 1 - 1/4 = 3/4$

Step 4: Value $= 1 / (3/4) = 4/3$

Answer: A 4/3

Rough work / extra work

Q10. L03 small angle

A circular disc of radius 10 cm subtends 0.04 rad at the eye. Using the small-angle approximation, the diameter of the disc as seen is about

2 marks

A 0.2 cm

B 0.4 cm

C 2 cm

D 4 cm

Solution: Q10. L03 small angle

$$\theta \text{ (rad)} \approx \text{size} / \text{distance}$$

Step 1: Small-angle: $\theta \approx$ (linear size) / distance

Step 2: Here distance = radius to eye = 10 cm, $\theta = 0.04$ rad

Step 3: diameter $\approx \theta \times$ distance $= 0.04 \times 10 = 0.4$ cm

Answer: B 0.4 cm

Use diameter (full size), not radius of the disc, when the question asks for diameter.

Q11. L04 max-min

The maximum value of $5 \sin \theta + 12 \cos \theta - 7$ is

2 marks

A 6

B 13

C 20

D -7

Solution: Q11. L04 max-min

$$y_{max} \text{ of } a\sin\theta + b\cos\theta = \sqrt{(a^2 + b^2)}$$

Step 1: For a sin θ + b cos θ, ymax = $\sqrt{(a^2 + b^2)}$

Step 2: $\sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$

Step 3: So max of (5 sin θ + 12 cos θ − 7) = 13 − 7 = 6

Answer: A 6

Do not write ymax = 5 + 12. Sin and cos cannot both be 1 at once.

Rough work / extra work

Q12. L04 max-min

The range of $f(\theta) = 2 + 3\sin\theta + 4\cos\theta$ is

2 marks

A $[-3, 7]$

B $[-5, 5]$

C $[2, 7]$

D $[-1, 5]$

Solution: Q12. L04 max-min

$$a\sin\theta + b\cos\theta \in [-\sqrt{(a^2+b^2)}, +\sqrt{(a^2+b^2)}]$$

Step 1: $3\sin\theta + 4\cos\theta$ lies between $-\sqrt{(9+16)}$ and $+\sqrt{(9+16)}$, i.e. $[-5, 5]$

Step 2: Add the constant 2: range of $f = [2 - 5, 2 + 5] = [-3, 7]$

Answer: A $[-3, 7]$

Q13. L04 phase

A particle has coordinates $x = 3\cos(\omega t)$ and $y = 4\sin(\omega t)$. Its maximum distance from the origin is

2 marks

A 3

B 4

C 5

D 7

Solution: Q13. L04 phase

Step 1: Eliminate t: $x^2/9 + y^2/16 = 1$ (ellipse, axes-aligned)

Step 2: Distance from origin is largest along the major axis

Step 3: Semi-major axis = 4 (along y)

Step 4: Maximum distance = 4

Answer: B 4

Do not write $\sqrt{(3^2+4^2)} = 5$ — that corner is outside the ellipse.

Q14. L05 phase

$y_1 = 4\sin(\omega t)$ and $y_2 = 3\sin(\omega t + \pi/2)$. The amplitude of $y_1 + y_2$ is

2 marks

A 5

B 7

C 1

D $\sqrt{7}$

Solution: Q14. L05 phase

$$\varphi = 90^\circ \rightarrow R = \sqrt{(A_1^2 + A_2^2)}$$

Step 1: Phase difference $\varphi = \pi/2$

Step 2: Resultant amplitude $R = \sqrt{(A_1^2 + A_2^2 + 2\,A_1\,A_2 \cos\varphi)}$

Step 3: $\cos(\pi/2) = 0 \rightarrow R = \sqrt{(4^2 + 3^2)} = \sqrt{(16 + 9)} = 5$

Step 4: Same as perpendicular phasors (our board method)

Answer: A 5

Do not add 4 + 3 = 7 unless the waves are exactly in phase.

Rough work / extra work

Q15. L05 AP

If the pth, qth and rth terms of an A.P. are a, b and c respectively, then a(q – r) + b(r – p) + c(p – q) equals

2 marks

A 0

B 1

C a + b + c

D pqr

Solution: Q15. L05 AP

Step 1: Write $a = A + (p - 1)D$, $b = A + (q - 1)D$, $c = A + (r - 1)D$

Step 2: Put into $a(q - r) + b(r - p) + c(p - q)$

Step 3: The A terms cancel: $A[(q-r)+(r-p)+(p-q)] = 0$

Step 4: The D terms also cancel identically → whole expression = 0

Answer: A 0

Q16. L06 AP

The sum of n terms of an A.P. is $3n^2 + 5n$. Its 10th term is

2 marks

A 32

B 59

C 62

D 65

Solution: Q16. L06 AP

$$T_n = S_n - S_{n-1}$$

Step 1: Class formula: $T_n = S_n - S_{n-1}$

Step 2: $S_{10} = 3(100) + 5(10) = 300 + 50 = 350$

Step 3: $S_9 = 3(81) + 5(9) = 243 + 45 = 288$

Step 4: $T_{10} = 350 - 288 = 62$

Answer: C 62

Rough work / extra work

Q17. L06 GP

In a G.P., the 3rd term is 24 and the 6th term is 192. The first term is

2 marks

A 3

B 6

C 8

D 12

Solution: Q17. L06 GP

Step 1: $ar^2 = 24$, $ar^5 = 192$

Step 2: Divide: $(ar^5)/(ar^2) = r^3 = 192/24 = 8 \rightarrow r = 2$

Step 3: $ar^2 = a \cdot 4 = 24 \rightarrow a = 6$

Answer: B 6

Q18. L07 GP

A ball rebounds to 4/5 of the height from which it falls. If it falls from 10 m, the total distance travelled before coming to rest is

2 marks

A 50 m

B 90 m

C 100 m

D 18 m

Solution: Q18. L07 GP

$$\text{Total} = h + 2 \cdot (h\,r)/(1 - r)$$

Step 1: First fall = 10 m (only once)

Step 2: After that: up + down form an infinite GP with first up = 10 × (4/5) = 8

Step 3: Sum of all ups and downs after first fall = 2 × [8 / (1 − 4/5)] = 2 × 8 / (1/5) = 80

Step 4: Total distance = 10 + 80 = 90 m

Answer: B 90 m

Do not forget the first fall is counted only once.

Q19. L06 straight line

The lines $2x - 3y + 6 = 0$ and $4x - 6y - 5 = 0$ are

2 marks

A coincident
B intersecting at one point
C parallel and distinct
D perpendicular

Solution: Q19. L06 straight line

Step 1: Slope of first: $2x - 3y + 6 = 0 \rightarrow y = (2/3)x + 2 \rightarrow m_1 = 2/3$

Step 2: Slope of second: $4x - 6y - 5 = 0 \rightarrow y = (2/3)x - 5/6 \rightarrow m_2 = 2/3$

Step 3: Same slope, but second is not a scalar multiple of the first (constants 6 vs −5)

Step 4: Therefore parallel and distinct

Answer: C parallel and distinct

Same slope alone is not enough for coincident — check the constant term too.

Rough work / extra work

Q20. L07 straight line

The equation of the line through (2, −3) and parallel to $3x + 4y - 7 = 0$ is

2 marks

A $3x + 4y + 6 = 0$

B $3x + 4y - 6 = 0$

C $4x - 3y - 17 = 0$

D $3x - 4y + 6 = 0$

Solution: Q20. L07 straight line

Step 1: Parallel line keeps the same x and y coefficients: $3x + 4y + k = 0$

Step 2: It passes through $(2, -3)$: $3(2) + 4(-3) + k = 0$

Step 3: $6 - 12 + k = 0 \rightarrow k = 6$

Step 4: Equation: $3x + 4y + 6 = 0$

Answer: A $3x + 4y + 6 = 0$

Q21. L08 graphs

A particle starts from rest with constant acceleration and covers 8 m in the first 2 s. Distance covered in the next 2 s is

2 marks

A 8 m

B 16 m

C 24 m

D 32 m

Solution: Q21. L08 graphs

$$s = ut + \tfrac{1}{2} a t^2$$

Step 1: $u = 0$, $s = ut + \tfrac{1}{2} a t^2 \rightarrow 8 = \tfrac{1}{2} a (2)^2 \rightarrow 8 = 2a \rightarrow a = 4\ m/s^2$

Step 2: Distance in first 4 s: $s_4 = \tfrac{1}{2}(4)(4)^2 = 2 \times 16 = 32$ m

Step 3: Next 2 s = $s_4 - s_2 = 32 - 8 = 24$ m

Answer: C 24 m

Rough work / extra work

Q22. L08 graphs

On a straight-line v–t graph, velocity falls from 10 m/s to 0 in 5 s. Displacement in this interval is

2 marks

A 10 m

B 25 m

C 50 m

D 0 m

Solution: Q22. L08 graphs

displacement = area under v–t graph

Step 1: On a v–t graph, displacement = area under the graph

Step 2: Here the graph is a straight triangle: base 5 s, height 10 m/s

Step 3: Area = ½ × 5 × 10 = 25 m

Answer: B 25 m

Q23. L08 graphs

If $a = -2\ m/s^2$ (constant) and initial velocity is $+10$ m/s, the particle comes to rest after

2 marks

A 2 s

B 5 s

C 10 s

D 20 s

Solution: Q23. L08 graphs

$$v = u + a\,t$$

Step 1: At rest: $v = 0$

Step 2: Use $v = u + a\,t$

Step 3: $0 = 10 + (-2)t \rightarrow 2t = 10 \rightarrow t = 5\ s$

Answer: B 5 s

Q24. L09 circle

For the circle $x^2 + y^2 - 4x + 6y - 3 = 0$, the centre and radius are

2 marks

A $(2, -3), 4$

B $(-2, 3), 4$

C $(2, -3), 2$

D $(2, 3), 4$

Solution: Q24. L09 circle

centre $(-g, -f)$, $r = \sqrt{(g^2 + f^2 - c)}$

Step 1: General form: $x^2 + y^2 + 2gx + 2fy + c = 0$

Step 2: Here $2g = -4 \rightarrow g = -2$; $2f = 6 \rightarrow f = 3$; $c = -3$

Step 3: Centre $= (-g, -f) = (2, -3)$

Step 4: Radius $= \sqrt{(g^2 + f^2 - c)} = \sqrt{(4 + 9 - (-3))} = \sqrt{16} = 4$

Answer: A $(2, -3), 4$

Rough work / extra work

Q25. L09 ellipse

For the ellipse $x^2/16 + y^2/9 = 1$, the distance between the foci is

2 marks

A $\sqrt{7}$

B $2\sqrt{7}$

C 5

D $\sqrt{5}$

Solution: Q25. L09 ellipse

foci distance = 2ae

Step 1: $a^2 = 16 \rightarrow a = 4$; $b^2 = 9 \rightarrow b = 3$ (a > b, major along x)

Step 2: $e = \sqrt{(1 - b^2/a^2)} = \sqrt{(1 - 9/16)} = \sqrt{(7/16)} = \sqrt{7} / 4$

Step 3: Distance between foci = $2ae = 2 \cdot 4 \cdot (\sqrt{7}/4) = 2\sqrt{7}$

Answer: B $2\sqrt{7}$

Q26. L09 exponential

If $y = e^{3x} + e^{-3x}$, then $d^2y/dx^2 - 9y$ equals

2 marks

A 0

B $18e^{3x}$

C $-18e^{-3x}$

D 9

Solution: Q26. L09 exponential

Step 1: $dy/dx = 3\,e^{3x} - 3\,e^{-3x}$

Step 2: $d^2y/dx^2 = 9\,e^{3x} + 9\,e^{-3x} = 9(e^{3x} + e^{-3x}) = 9y$

Step 3: Therefore $d^2y/dx^2 - 9y = 0$

Answer: A 0

Rough work / extra work

Q27. L09 differentiation

If $y = \sqrt{(2x + 3)}$, then dy/dx at $x = 3$ equals

2 marks

A $1/\sqrt{3}$

B $1/\sqrt{9}$

C $1/3$

D $\sqrt{3}$

Solution: Q27. L09 differentiation

$$\frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \cdot \frac{du}{dx}$$

Step 1: Write $y = (2x + 3)^{1/2}$

Step 2: Chain rule: $\frac{dy}{dx} = \frac{1}{2}(2x + 3)^{-1/2} \cdot 2 = \frac{1}{\sqrt{2x + 3}}$

Step 3: At $x = 3$: $\frac{1}{\sqrt{6 + 3}} = \frac{1}{\sqrt{9}} = \frac{1}{3}$

Answer: C 1/3

Q28. L10 rate of change

The volume of a sphere increases at 4π cm^3/s when the radius is 2 cm. The rate of increase of the radius is

2 marks

A 1/4 cm/s

B 1/2 cm/s

C 1 cm/s

D 2 cm/s

Solution: Q28. L10 rate of change

$$dV/dt = 4\pi r^2 \, dr/dt$$

Step 1: $V = (4/3)\pi r^3 \rightarrow dV/dt = 4\pi r^2 (dr/dt)$

Step 2: Given $dV/dt = 4\pi$ when $r = 2$

Step 3: $4\pi = 4\pi (2)^2 (dr/dt) = 4\pi \cdot 4 \cdot (dr/dt)$

Step 4: $1 = 4 (dr/dt) \rightarrow dr/dt = 1/4$ cm/s

Answer: A 1/4 cm/s

Q29. L10 differentiation

If $y = (x^2 + 1)/(x^2 - 1)$ with $|x| \neq 1$, then dy/dx equals

2 marks

A $-4x/(x^2 - 1)^2$

B $4x/(x^2 - 1)^2$

C $-2x/(x^2 - 1)^2$

D $2x/(x^2 - 1)$

Solution: Q29. L10 differentiation

$$(u/v)' = (v\,u' - u\,v') / v^2$$

Step 1: Quotient rule: $(u/v)' = (v\,u' - u\,v') / v^2$

Step 2: $u = x^2 + 1$, $u' = 2x$; $v = x^2 - 1$, $v' = 2x$

Step 3: Numerator $= (x^2 - 1)(2x) - (x^2 + 1)(2x) = 2x[(x^2 - 1) - (x^2 + 1)] = 2x(-2) = -4x$

Step 4: $dy/dx = -4x / (x^2 - 1)^2$

Answer: A $-4x/(x^2 - 1)^2$

Keep the order v u' − u v' — swapping the sign is a common trap.

Rough work / extra work

Q30. L10 product / chain

If $y = \sin(x^2)$, then dy/dx equals

2 marks

A $\cos(x^2)$

B $2x \cos(x^2)$

C $2x \sin(x^2)$

D $\cos(2x)$

Solution: Q30. L10 product / chain

$$\frac{d}{dx}[\sin u] = \cos u \cdot \frac{du}{dx}$$

Step 1: Outer function $\sin(\cdot)$, inner function x^2

Step 2: Chain rule: $\frac{dy}{dx} = \cos(x^2) \cdot \frac{d(x^2)}{dx}$

Step 3: $\frac{d(x^2)}{dx} = 2x$

Step 4: $\frac{dy}{dx} = 2x\cos(x^2)$

Answer: B $2x\cos(x^2)$

Do not forget to multiply by the derivative of the inside (2x).

Paper discussion lock lines

Every question: write the class formula first, then substitute

ASTC / $\sqrt{(a^2+b^2)}$ / $T_n = S_n - S_{n-1}$ / $\ell = r\theta$ / quotient–chain — same as boards

Read options after you finish; do not reverse-engineer from choices

Re-attempt any wrong question tomorrow without looking at the solution slide.

Practice

1. Re-do Q1–Q10 from memory (ratios, ASTC, compound, small angle).

2. Re-do Q11–Q20 (max–min, phase, AP/GP, lines).

3. Re-do Q21–Q30 (graphs, circle, ellipse, differentiation).

4. Mark every trap in orange in your notebook.

If a method feels new, go back to that lecture slide — we stay with class methods only.

Rough work / extra work